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T11 – 05

Assignment No. 7

Study Assignment

1. Bluetooth Technology

Introduction:

Bluetooth is a wireless communication technology that enables short-range data exchange between devices using radio waves. It operates in the 2.4 GHz ISM (Industrial, Scientific, and Medical) band and is widely used for connecting mobile phones, computers, headsets, speakers, and various smart devices.

Developed by Ericsson in 1994 and later formalized by the Bluetooth Special Interest Group (SIG) in 1998, Bluetooth has become a standard for low-power, low-cost wireless communication.

Working Principle

Bluetooth uses a technique called Frequency Hopping Spread Spectrum (FHSS), which:

- Divides the 2.4 GHz band into 79 channels (in most countries).
- Devices hop between channels up to 1600 times per second, reducing interference and improving security.
- Data is transmitted in small packets and shared across multiple devices within a short range (typically 10 meters, extendable up to 100 meters with Bluetooth Class 1).

Bluetooth Architecture

Bluetooth communication involves a Master-Slave relationship:

- Master Device: Initiates and controls communication.

- Slave Devices: Respond to the master's commands.

Piconet:

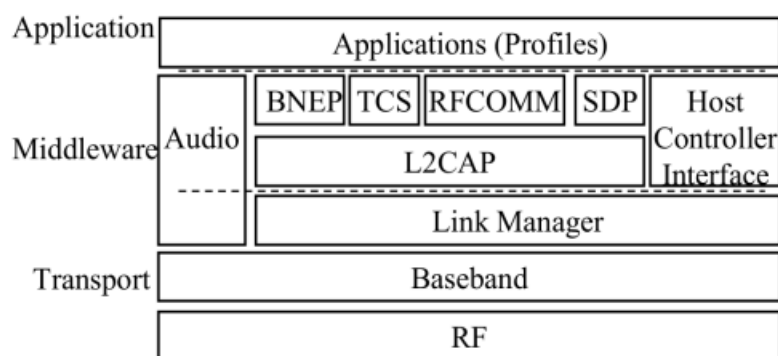
- A small network formed by one master and up to 7 active slave devices.
- Devices share the master's clock and frequency hopping pattern.

Scatternet:

- Multiple interconnected piconets.
- A device can act as a slave in one piconet and a master in another, enabling more complex networks.

Bluetooth Protocol Stack

Bluetooth Protocol Stack



The Bluetooth protocol stack consists of several layers, similar to the OSI model:

1. Radio Layer:
 - Physical layer responsible for modulation and transmission.
2. Baseband Layer:
 - Handles error correction, packet formatting, and addressing.
3. Link Manager Protocol (LMP):
 - Establishes and manages connections (authentication, encryption, power control).
4. Host Controller Interface (HCI):

- Acts as a bridge between hardware (controller) and software (host).
5. Logical Link Control and Adaptation Protocol (L2CAP):
- Handles multiplexing of data, segmentation, and reassembly.
6. RFCOMM:
- Emulates serial ports, enabling legacy applications to use Bluetooth.
7. Service Discovery Protocol (SDP):
- Allows devices to discover services provided by other Bluetooth devices.

Bluetooth Versions and Enhancements

Bluetooth technology has evolved over time with various versions introducing new capabilities:

Version	Year	Key Features
1.0 – 1.2	1999	Basic functionality, FHSS
2.0 + EDR	2004	Enhanced Data Rate (3 Mbps)
3.0 + HS	2009	High Speed (24 Mbps using Wi-Fi)
4.0	2010	Introduction of Bluetooth Low Energy (BLE)
4.1	2013	Better coexistence with LTE, improved connection
4.2	2014	Increased privacy, IPv6 support
5.0	2016	Longer range (240m), faster speed (2 Mbps), larger broadcast messaging
5.1	2019	Direction finding (angle of arrival/departure)

Version	Year	Key Features
5.2	2020	LE Audio, Isochronous Channels
5.3	2021	Enhanced periodic advertising, lower power
5.4	2023	Optimizations for large-scale Bluetooth networks (like in industrial IoT)

Applications of Bluetooth

Consumer Electronics

- Wireless headphones, earbuds, speakers
- Gaming controllers
- Smart TVs, remotes

Healthcare

- Heart rate monitors
- Glucose sensors
- Fitness trackers

IoT and Smart Homes

- Smart bulbs, thermostats
- Smart locks, doorbells
- Mesh networking for home automation

Automotive

- Hands-free calling
- Audio streaming
- In-vehicle data exchange

Industrial and Retail

- Asset tracking using Bluetooth beacons
- Inventory management

- Wearable devices for worker safety
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Bluetooth Security Features

Bluetooth employs several security measures:

- Authentication: Verifying the identity of communicating devices.
- Authorization: Ensuring only permitted devices can access services.
- Encryption: Protects the confidentiality of the transmitted data.
- Pairing and Bonding: Devices exchange keys during the pairing process, and store them for future secure connections.

Recent versions like Bluetooth 4.2 and above also support LE Secure Connections, enhancing security using Elliptic Curve Diffie-Hellman (ECDH).

Advantages of Bluetooth

- Low power consumption (especially BLE).
 - Cost-effective.
 - Universally adopted and compatible.
 - Operates without internet or complex infrastructure.
 - Supports voice and data transfer simultaneously.
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Limitations

- Limited data range compared to Wi-Fi.
- Potential for interference in the crowded 2.4 GHz band.
- Security concerns if older versions are used or pairing is not managed properly.
- Lower data transfer speed than Wi-Fi or 5G.

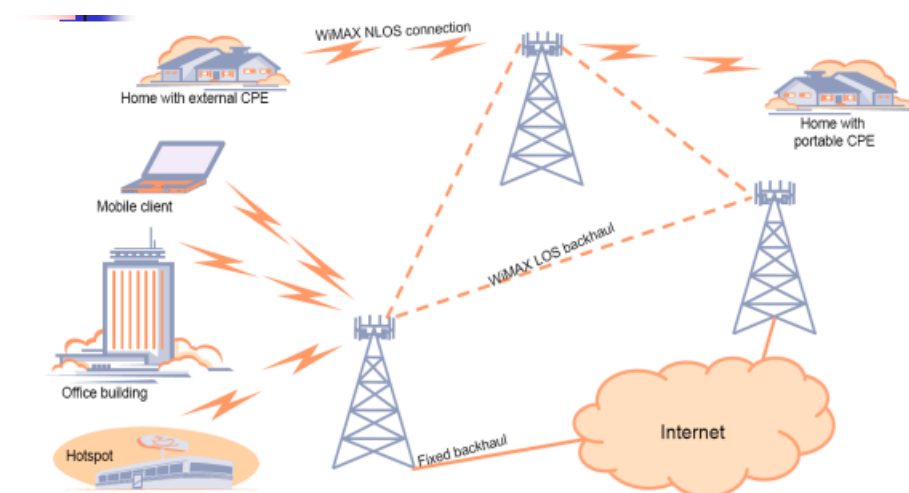
2. Wireless LAN Technology

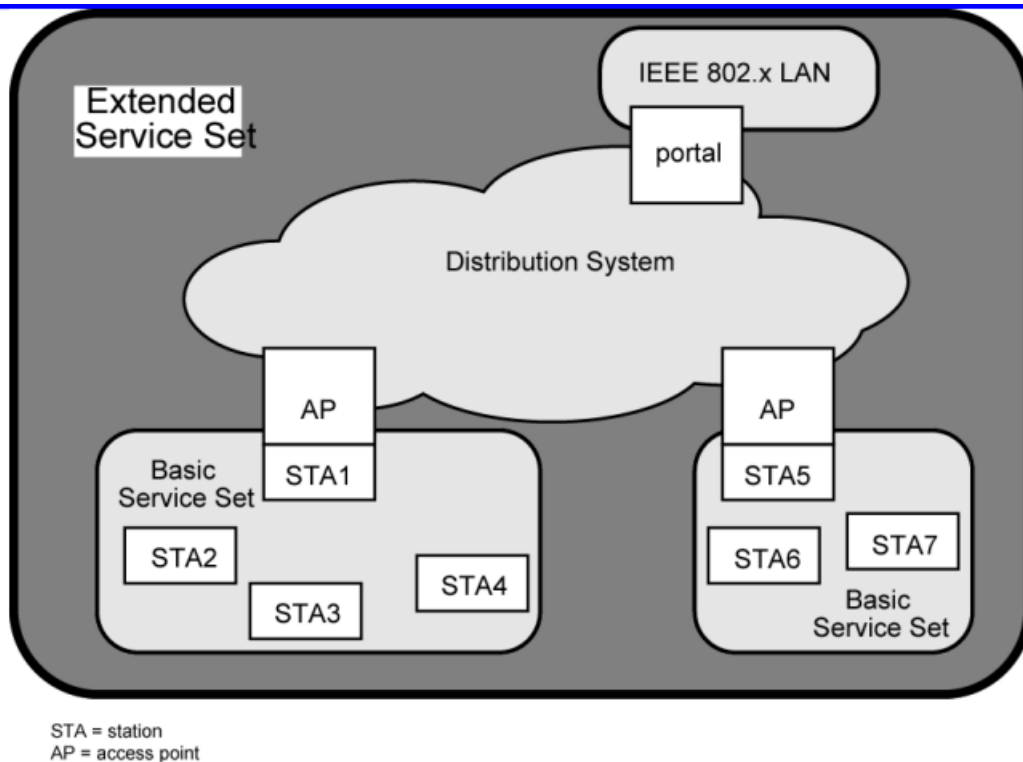
Introduction:

A Wireless Local Area Network (WLAN) is a type of local area network that uses radio frequency (RF) technology to allow devices to communicate wirelessly within a limited geographical area such as a home, office, campus, or public hotspot. WLANs are based on IEEE 802.11 standards, commonly known as Wi-Fi.

Unlike traditional wired LANs, WLANs provide mobility, flexibility, and convenience, making them essential in modern computing environments.

Basic Components of WLAN





1. Access Point (AP):

- A networking hardware device that connects wireless devices to a wired network.
- Acts as a central transmitter and receiver of wireless radio signals.

2. Wireless Stations (Clients):

- Devices like laptops, smartphones, tablets, or IoT devices equipped with Wi-Fi adapters.

3. Wireless Network Interface Card (NIC):

- Installed in each client device, enabling it to send and receive wireless signals.

4. Distribution System (DS):

- Connects multiple access points to each other or to the backbone network (LAN).

How WLAN Works

- WLANs use radio waves in the 2.4 GHz or 5 GHz frequency bands.
 - An Access Point broadcasts a signal (SSID), which devices detect and connect to.
 - Once connected, data packets are transmitted wirelessly using protocols like CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).
 - Data can be sent either directly (Ad hoc) or via the access point (Infrastructure mode).
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WLAN Architecture

- ◆ Infrastructure Mode
 - Most common setup.
 - Clients communicate through a central access point.
 - Enables integration with the wired LAN and the internet.
 - ◆ Ad Hoc Mode (Peer-to-Peer)
 - Devices communicate directly without an access point.
 - Suitable for temporary or small networks.
 - ◆ Mesh Networks
 - Multiple access points communicate and route data dynamically.
 - Offers wider coverage and redundancy.
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IEEE 802.11 Standards (Wi-Fi Versions)

Standard	Frequency	Max Speed	Range	Description
802.11b	2.4 GHz	11 Mbps	~100m	First widely used Wi-Fi standard
802.11a	5 GHz	54 Mbps	~50m	Faster but shorter range

Standard	Frequency	Max Speed	Range	Description
802.11g	2.4 GHz	54 Mbps	~100m	Combines speed of 802.11a with range of 802.11b
802.11n	2.4/5 GHz	600 Mbps	~250m	MIMO technology introduced
802.11ac	5 GHz	1.3 Gbps	~100m	Wider channels, beamforming
802.11ax (Wi-Fi 6)	2.4/5/6 GHz	10+ Gbps	~300m	Increased efficiency and capacity
802.11be (Wi-Fi 7)	2.4/5/6 GHz	30+ Gbps	~300m+	Ultra high throughput (still emerging)

WLAN Security Mechanisms

WLANs are vulnerable to eavesdropping, spoofing, and unauthorized access. Security protocols are essential to ensure safe communication:

◆ WEP (Wired Equivalent Privacy)

- First encryption protocol.
- Weak and outdated due to vulnerabilities.

◆ WPA (Wi-Fi Protected Access)

- Improved encryption and integrity over WEP.

◆ WPA2 (Most common)

- Uses AES encryption (Advanced Encryption Standard).
- Secure and widely adopted.

◆ WPA3

- Introduced in 2018.
- Enhanced security with Simultaneous Authentication of Equals (SAE).

- Better protection against brute-force attacks and public Wi-Fi threats.
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Applications of WLAN

1. Home Networking:
 - Internet access for smartphones, smart TVs, and IoT devices.
 2. Enterprise Networks:
 - Mobility for employees, easy device management.
 3. Education:
 - Campus-wide access for e-learning.
 4. Public Hotspots:
 - Airports, cafes, and libraries provide internet access.
 5. Healthcare:
 - Real-time access to patient records, monitoring systems.
 6. Industrial IoT:
 - Smart factory systems and sensor networks.
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Advantages of WLAN

- No cabling: Reduces cost and clutter.
 - Mobility: Users can move freely within coverage area.
 - Quick installation: No need to lay physical wires.
 - Scalability: Easy to add new users/devices.
 - Device compatibility: Supported by nearly all modern devices.
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Limitations

- Limited range: Coverage is less compared to wired LANs.
- Security risks: Wireless signals are easier to intercept.

- Interference: From other wireless devices like microwaves, Bluetooth.
- Speed variability: Performance can drop with distance or congestion.
- Power consumption: Portable devices may drain battery faster due to Wi-Fi usage.

Enhancements in WLAN Technology

- MIMO (Multiple Input, Multiple Output): Boosts data rates by using multiple antennas.
- Beamforming: Directs signal toward the client to improve speed and reliability.
- OFDMA (Orthogonal Frequency Division Multiple Access): Efficient use of spectrum by allowing simultaneous communication with multiple devices (used in Wi-Fi 6).
- TWT (Target Wake Time): Reduces power usage in IoT and mobile devices by scheduling when devices should wake up to communicate.

WLAN vs Other Wireless Technologies

Technology	Use Case	Range	Speed	Frequency
WLAN (Wi-Fi)	Internet/data	100-300m	Up to 10+ Gbps	2.4/5/6 GHz
Bluetooth	Peripheral connections	~10m	Up to 3 Mbps (classic), 2 Mbps (BLE)	2.4 GHz
ZigBee	Low-power IoT	~100m	250 kbps	2.4 GHz
Cellular (4G/5G)	Mobile networks	Kilometers	1 Gbps (4G), 10 Gbps+ (5G)	Licensed spectrum

